ADVANCED OBJECT ORIENTED PROGRAMMING-25CS2205E

Skill-1

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Section-4

Question 1

Create a class Student with student ID, name and age. Create three student objects and display their details.

Code:

```java
class Student{
    int id;
    String name;
    int age;

    void display(){
        System.out.println("id="+id);
        System.out.println("name="+name);
        System.out.println("age="+age);
    }
}
    public class Ex1{
    Run | Debug
    public static void main(String[]args){
        Student c=new Student();
        c.id=03;
        c.name="bhavana";
        c.age=18;
        c.display();
    }
}
```

Output:

```
Data\Roaming\Code\U
id=3
name=bhavana
age=18
PS D:\AOOP>
```

Question 2

Create an Employee class with employee ID, name and salary. Initialize the values using a constructor and display employee details.

Code:

```java
Ex2.java > Ex2 > Employee
public class Ex2 {
    static class Employee{
        int id;
        String name;
        int salary;
        Employee(int i,String n,int s){
            id=i;
            name=n;
            salary=s;
        }
        void display(){
            System.out.println("id="+id);
            System.out.println("name="+name);
            System.out.println("salary="+salary);

        }
    }
    Run | Debug
    public static void main(String[]args){
        Employee e=new Employee(i: 34,n: "bhavana",s: 5896);
        e.display();
    }
}

```

Output:

```
PS D:\AOOP>  & 'C:\U
p=transport=dt_socke
Data\Roaming\Code\Us
id=34
name=bhavana
salary=5896
PS D:\AOOP>
```

Question 3

Create a BankAccount class with account number and balance. Implement deposit and withdrawal methods.

Code:



Output:

```
PS D:\AOOP>  & 'C:\Users\
p=transport=dt_socket,ser
Data\Roaming\Code\User\wo
deposited = 2000.0
money withdrawl = 5000.0
total balance = 5000.0
account number1234
PS D:\AOOP>
```

Question 4

Create a Rectangle class with length and width. Calculate and display its area and perimeter.

Code:

```java
Ex4.java > Ex4 > Rectangle
public class Ex4 {
    static class Rectangle {
        int length;
        int width;

        Rectangle(int l, int w) {
            length = l;
            width = w;
        }

        void display() {
            int area = length * width;
            int perimeter = 2 * (length + width);

            System.out.println("Area = " + area);
            System.out.println("Perimeter = " + perimeter);
        }
        }
        Run | Debug
        public static void main(String[] args) {
            Rectangle r = new Rectangle(l: 10, w: 5);
            r.display();
        }

    }

```

Output:

```
Area = 50
Perimeter = 30
PS D:\AOOP>
```

Question 5

Create a Book class with ISBN, title and price. Initialize the values using a parameterized constructor.

Code:

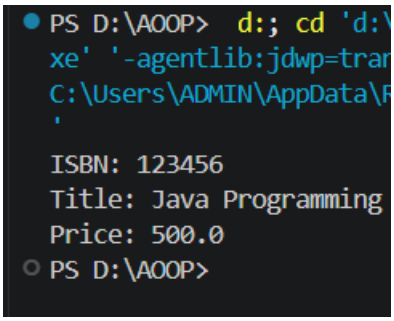
```java
public class Ex5 {
static class Book {
    String ISBN;
    String title;
    double price;

    // Parameterized constructor
    Book(String i, String t, double p) {
        ISBN = i;
        title = t;
        price = p;
    }

    void display() {
        System.out.println("ISBN: " + ISBN);
        System.out.println("Title: " + title);
        System.out.println("Price: " + price);
    }
}
    Run | Debug
    public static void main(String[] args) {
        Book b = new Book(i: "123456", t: "Java Programming", p: 500);

        b.display();
    }
}
```

output:


PS D:\AOOP> d:; cd 'd:\
xe' '-agentlib:jdwp=tran
C:\Users\ADMIN\AppData\R
'
ISBN: 123456
Title: Java Programming
Price: 500.0
PS D:\AOOP>